

Spectral Analysis of the Groundwater Levels in a Suburban vs Urban Region: An Example from Suffolk and Queens Counties in Long Island, NY

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Abstract

The supplied water in Long Island, NY originates from the underground reserves available throughout the island. Factors such as the installation of sanitary, storm-sewer systems, and large scale groundwater pumpage have been affecting the amount of water in the reserves since the area first began being developed. We have used data retrieved from the United States Geological Survey (USGS), monitoring stations for the water depth levels in order to analyze the long-term correlation between the groundwater in Queens and Suffolk County and their related periodicities. Groundwater measurements were retrieved from two USGS stations located in Queens and Suffolk County, both within the Upper Glacial Aquifer. The data was imported and analyzed in KoNstanz Information MinEr (KNIME) to evaluate any possible missing values and applied different data analytics techniques. Groundwater time series analysis was performed by detrending the time series, and further separation of the various time scale components (long, seasonal, and short term component). With this information, we determined the groundwater period in urban Queens versus suburban Suffolk county.



Figure 1: Study area and USGS data collection sites (404416073491802, 405743072425701 for Queens and Suffolk county respectively) are projected on a satellite image produced in Google Earth. The dashed line represents the groundwater divide of the watershed in Long Island, NY (Franke & McClymonds, 1972).

Introduction

The supplied water in Long Island, NY originates from the underground reserves available throughout the island. The groundwater levels in Queens and Suffolk County (Figure 1). Factors such as the installation of sanitary, storm-sewer systems, and large scale groundwater pumpage have been affecting the amount of water in the reserves since the area first began being developed. The bedrock of both locations are similar due to Long Island’s glacial depositional origin. The population ratio is 2:1 with Queens in the lead. Suffolk and Queens share an aquifer (Figure 1), however, the residents of Queens primarily depend on the metropolitan water source located in Upstate New York. Another factor to consider is the area and volume of the aquifer at each location. These two areas have been studied in order to monitor the periodicity of groundwater in each location (Figure 1). We have used data retrieved from the United States Geological Survey (USGS) monitoring stations for the water depth levels in order to analyze the long-term correlation between the groundwater in Queens and Suffolk County and their related periodicities.

Methods

- The data sets of daily groundwater levels were collected from USGS, ranging from March 3, 2014, to February 26, 2019. Groundwater measurements were retrieved from two USGS stations located in Queens and Suffolk County.
- The data was imported and analyzed in KoNstanz Information MinEr (KNIME) to evaluate any possible missing values and applied different data analytics techniques.
- The groundwater data was then detrended to convert the groundwater time series in a stationary time series in order to study the different periodicities at both locations.
- Applying the spectral analysis method, we determined the main periodicities for both groundwater time series.
- The spectral analysis has been conducted in R software using the time series analysis package (TSA package in R; Chan and Ripley, 2018) for the groundwater levels in Queens and Suffolk.
- Groundwater time series analysis was performed by detrending the time series, and further separation of the various time scale components (long, seasonal, and short-term component).
- With this information, we may compare the groundwater level periodicity in urban Queens versus suburban Suffolk county.

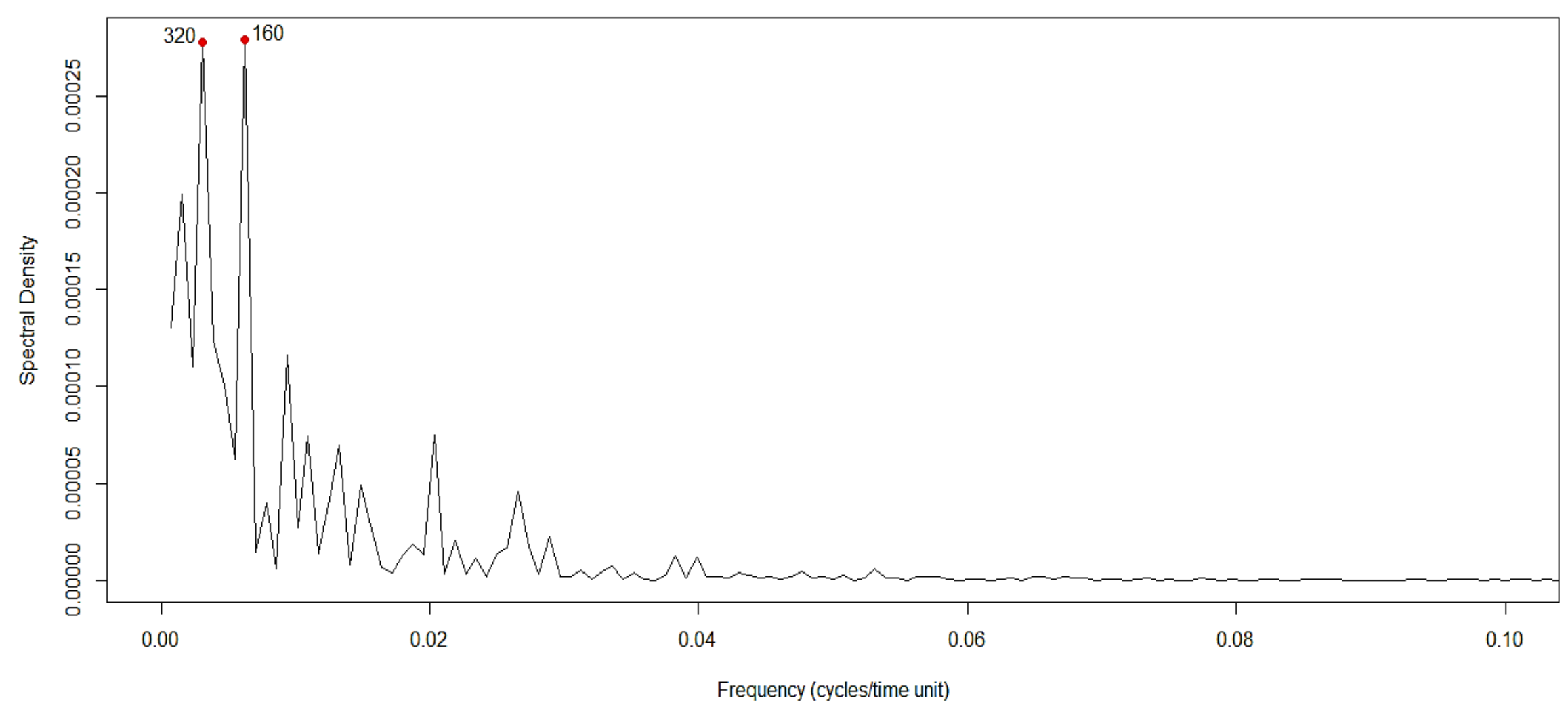
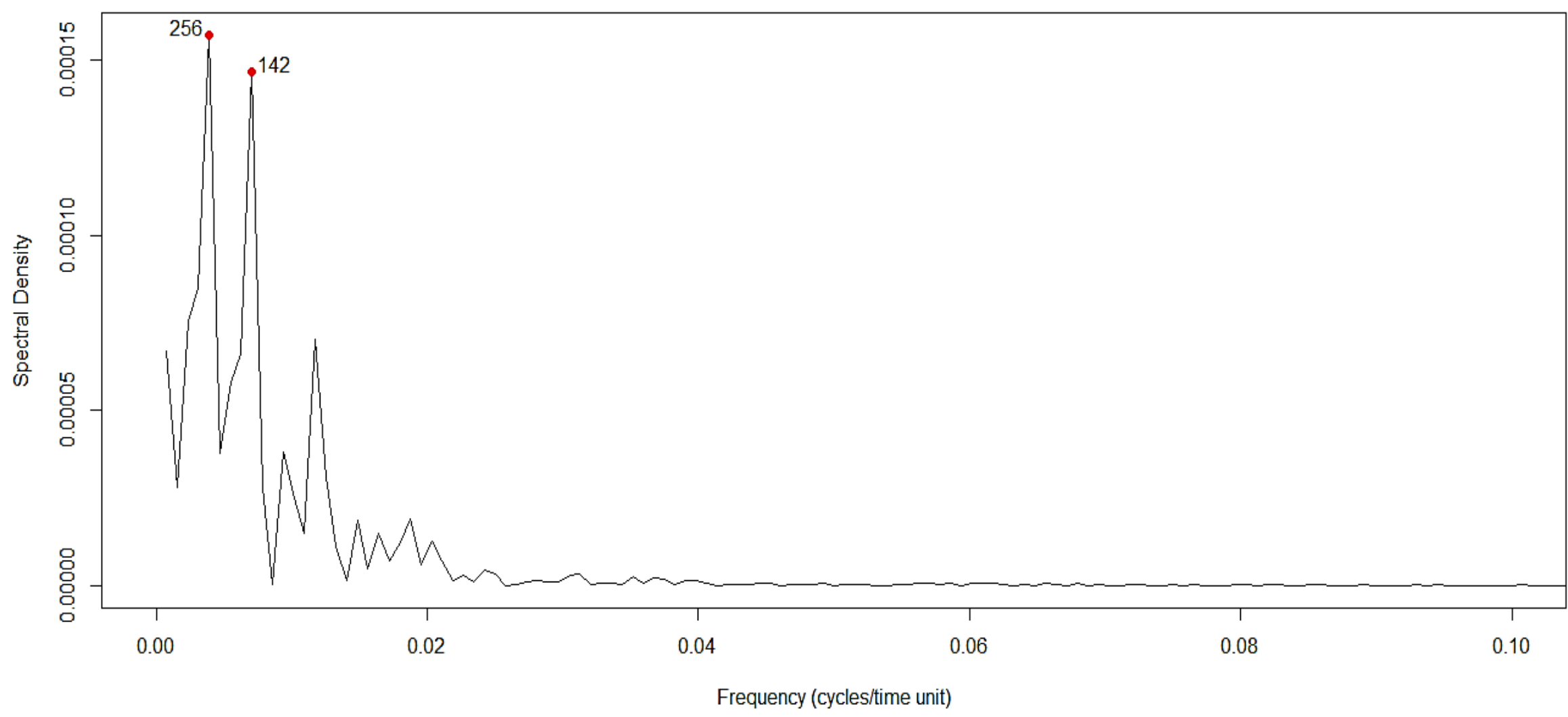


Figure 2: (Above) Periodograms showing the major cycles of groundwater for Queens (upper) and Suffolk region (lower). Periodicities (in days) are showing with a red bullet on each corresponding spectral density peak.

Results

- There are two peaks in Queens, day 256 (major), day 142 (minor)
- There are two peaks in Suffolk, day 320 and 160 (both major)
- There is a shorter periodicity in Queens than in Suffolk

Queens		
Frequency (cycles/time unit)	Spectral	Period (days)
0.00391	0.00016	256
0.00703	0.00015	142

Suffolk		
Frequency (cycles/time unit)	Spectral	Period (days)
0.00625	0.00028	160
0.00312	0.00028	320

Table 1: (Left) For the time period of 2015-2019, the spectral density, related frequency (cycles/time unit), and periods are presented for Suffolk county USGS station 405743072425701 and Queens county USGS station 404416073491802.

Discussion

- Groundwater recharges at a faster rate in Queens than Suffolk.
- Queens is less surrounded by seawater and more to land, while Suffolk is more making run-off into the ocean more likely to occur in Suffolk.
- Queens recharges faster than Suffolk, which contradicted our expected results since Queens is more urbanized suggesting water is less likely to percolate into the aquifer.
- Queens utilizes water from Upstate rather than the upper glacial aquifer, therefore, the groundwater under Queens is able to recharge at a quicker rate due to the lack of significant usage.
- Although the surface area of Suffolk is nearly twenty times the size of Queens’ surface area, the aquifer isn’t limited to county boundaries.
- Queens is adjacent to the other watersheds in the metropolitan area.

Conclusion

The spectral analysis of the groundwater level from the two USGS stations for the time period of 2015 through 2019 has shown two main spectral densities that correspond to the periods of 256 days and 142 days for Queens station, and the periods of 320 days and 160 days for Suffolk station.

References

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